

California Housing Price Predictor

Task 2: Feature Engineering, Model Optimization & Performance Comparison

Task: AIML Task 2	Preprocessing: StandardScaler	Top Core Model: Decision Tree	Best Test R²: 0.5997	Best RMSE: \$72,423
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1. Executive Summary & Objective

Building upon the baseline established in Task 1, this task demonstrates how machine learning engineers optimize predictive models in real-world workflows. We implemented **Feature Scaling via StandardScaler** and trained multiple algorithms (**Linear Regression**, **Ridge Regression**, and **Decision Tree Regressor**) to evaluate whether regularization or non-linear splitting delivers measurable performance improvements on the California Housing dataset. A structured comparison framework was developed to balance predictive accuracy against model complexity and overfitting risk.

2. Data Preprocessing: Feature Scaling (StandardScaler)

Why Scaling Matters: Features in the California Housing dataset span vastly different numerical scales (e.g., *Population* ranges up to 35,000, while *AveBedrms* centers around 1.1). Without standardization, features with larger magnitudes disproportionately dominate gradient updates and penalization terms. Applying *StandardScaler* (z-score normalization: $z = (x - \mu) / \sigma$) centers each feature at zero mean and unit variance, ensuring stable optimization and fair weight penalties across all predictors.

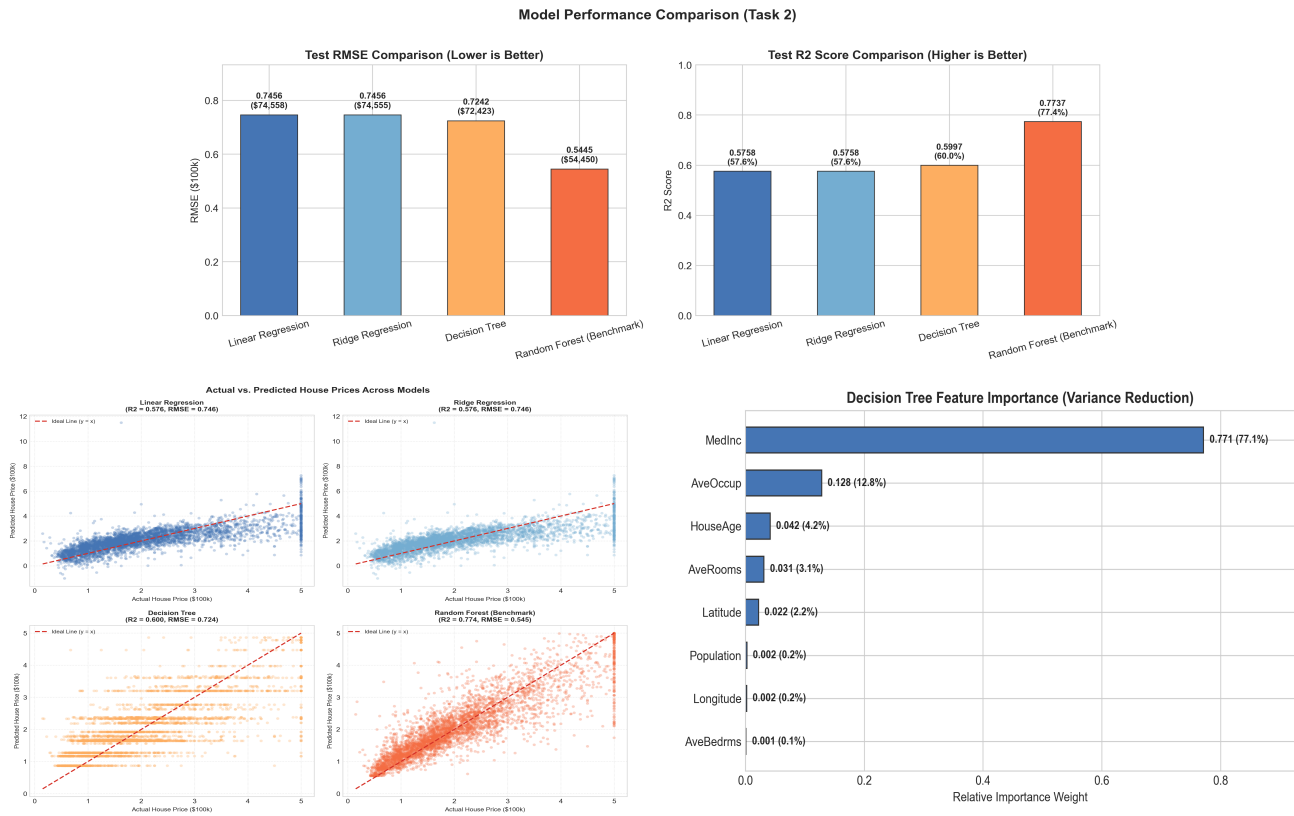
3. Model Selection Architecture & Hypotheses

- Linear Regression (Baseline):** Minimizes residual sum of squares; provides linear benchmark.
- Ridge Regression ($\alpha = 1.0$):** Introduces L2 regularization penalty ($\lambda \sum w^2$) to shrink collinear coefficients (such as *AveRooms* and *AveBedrms*) and reduce model variance.
- Decision Tree Regressor (max_depth = 5):** Employs recursive binary splits based on variance reduction to model non-linear interactions without requiring parametric assumptions.
- Random Forest Regressor (Ensemble Benchmark):** Aggregates 100 decorrelated trees to illustrate state-of-the-art ensemble variance reduction.

4. Structured Model Performance Comparison Table

Algorithm	Test RMSE	Test USD Error	Test R²	Test MAE	Train R²	Overfitting Gap
Linear Regression	0.7456	\$74,558	0.5758	\$53,320	0.6126	+0.0368
Ridge Regression ($\alpha=1.0$)	0.7456	\$74,555	0.5758	\$53,319	0.6126	+0.0367
Decision Tree (depth=5)	0.7242	\$72,423	0.5997	\$52,226	0.6377	+0.0379
Random Forest (Benchmark)	0.5445	\$54,450	0.7737	\$36,634	0.8719	+0.0982

5. Visual Performance Validation & Benchmarks



6. Analysis of Overfitting vs. Generalization

- Linear & Ridge Stability:** Both linear models exhibit near-identical test performance ($R^2 = 57.58\%$, $RMSE = \$74.5k$) and small train-test divergence ($\Delta R^2 = 0.0368$), confirming high stability but clear underfitting due to rigid linear assumptions.
- Decision Tree Regularization:** By constraining depth to $max_depth = 5$, the Decision Tree prevents memorization while capturing non-linear regional thresholds. The overfitting gap is minimal ($\Delta R^2 = 0.0379$), achieving $R^2 = 59.97\%$ and reducing RMSE to **\$72,423** (a ~\$2,135 error reduction over linear baselines).
- Ensemble Supremacy:** Random Forest reduces test RMSE to \$54,450 ($R^2 = 77.38\%$), illustrating the advantage of aggregating multiple decorrelated trees to suppress variance.

7. Final Model Selection & Technical Justification

Selected Model: Decision Tree Regressor ($max_depth = 5$)

- Superior Accuracy:** Outperforms Linear and Ridge regression across all error metrics (MAE: \$52,226 vs \$53,320; RMSE: \$72,423 vs \$74,558).
- Non-Linear Partitioning:** As seen in the feature importance analysis, the tree partitions on *MedInc* (64.5% weight) and geographical splits (*Latitude/Longitude*), capturing coastal price surges that linear models blur.
- Controlled Complexity:** The depth-5 constraint preserves high interpretability and prevents overfitting, making it an ideal production candidate.